

PHYSICS 1

UNIT 1: KINEMATICS

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Name: _____

How to Use This Packet

Each section below has two parts, in the Cornell note-taking format, plus a summary box.

Notes column (right, wide)

Fill this in during lecture. Definitions, formulas, and derivations are partly blanked — write the missing words, symbols, or steps as your instructor presents them. Worked examples give you the problem; you fill in every step of the solution.

Cue column (left, narrow)

These are the key questions and terms for the section. Use them right after class (or before a quiz) to test yourself: cover the notes column and see if you can answer each cue from memory.

Summary box

At the end of each section, write 2–3 sentences in your own words capturing the main idea. If you can't summarize it, you don't know it yet — go back and review.

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Chapter 1: Introduction

1.1 Physics: An Introduction

CUE / KEY QUESTIONS

Definition of physics?

2 pillars of empirical science

5 steps of the scientific method, in order

3 branches of physics + their focus

3 reasons to study physics

What Is Physics?

Physics is the branch of science concerned with the nature and properties of _____ and _____.

Two key goals of physics:

Physics Is Empirical — It Relies On:

The Scientific Method (write the 5 steps, in order)

Branches of Physics — fill in the focus area for each

Classical Mechanics:

Thermodynamics:

Electromagnetism:

Optics:

Modern Physics:

	Why Study Physics? (list 3 reasons)
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SUMMARY — in your own words, summarize this section in 2-3 sentences:

1.2 Physical Quantities and Units

CUE / KEY QUESTIONS

What is a physical quantity?

Name the 7 SI base units + symbols

One example of a derived unit

What is dimensional analysis used for?

Metric prefix for 10^9 ? For 10^{-3} ?

Physical Quantities

A physical quantity is a _____ property of the physical world (e.g., length, mass, time, temperature, current).

SI Base Units — fill in the missing symbols

Length (meter): _____ Mass (kilogram): _____ Time (second): _____

Electric current (ampere): _____ Temperature (kelvin): _____

Amount of substance (mole): _____ Luminous intensity (candela): _____

Metric Prefixes — fill in the power-of-ten factor

giga (G) = $10^{\text{_____}}$ mega (M) = $10^{\text{_____}}$ kilo (k) = $10^{\text{_____}}$

centi (c) = $10^{\text{_____}}$ milli (m) = $10^{\text{_____}}$ micro (μ) = $10^{\text{_____}}$

nano (n) = $10^{\text{_____}}$

Derived Units (write the formula/unit)

Speed = _____

Force (newton, N) = _____

Energy (joule, J) = _____

Dimensional Analysis

Rule: any valid physical formula must be _____.

Practice: circle the equations below that are dimensionally consistent

$x = vt$ $x = \frac{1}{2}at^2$ $t = \sqrt{2x/a}$

SUMMARY — in your own words, summarize this section in 2-3 sentences:

1.3 Accuracy, Precision, and Significant Figures

CUE / KEY QUESTIONS

Define accuracy

Define precision

Can a measurement be precise but not accurate?

Example?

5 rules for counting sig figs

Sig-fig rule:
multiplication/division vs.
addition/subtraction

Accuracy vs. Precision

Accuracy: how close a measured value is to the _____ value.

Precision: how close repeated measurements are to _____, regardless of the true value.

Example: true value = 9.8 m/s^2 . A measurement of 9.79 m/s^2 is _____ (accurate / precise).

Example: measurements of 9.65 , 9.66 , 9.65 m/s^2 are _____ even if not accurate.

Uncertainty

All measurements have uncertainty, written as: Measured Value $\pm$ _____

Three sources of uncertainty:

Rules for Counting Significant Figures — fill in the sig-fig count and the rule

$123 \rightarrow$ _____ sig figs (rule: _____)

$1002 \rightarrow$ _____ sig figs (rule: _____)

$0.0045 \rightarrow$ _____ sig figs (rule: _____)

$45.00 \rightarrow$ _____ sig figs (rule: _____)

$1500 \rightarrow$ _____ sig figs (ambiguous — use _____ notation to clarify)

Sig Figs in Calculations

Multiplication & Division: result has the same number of sig figs as the value with the _____ sig figs.

Addition & Subtraction: result has the same number of _____ places as the value with the fewest.

Worked Example

A tortoise travels at 2.51 cm/s for 12.23 s. How far does it go?

Raw calculation (no rounding):

Fewest sig figs among the given values:

Final answer, correctly rounded:

SUMMARY — in your own words, summarize this section in 2–3 sentences:

1.4 Approximation

CUE / KEY QUESTIONS

2 reasons approximation is useful

What is an order-of-magnitude estimate?

Mass of a car $\approx$? People in a stadium $\approx$?

Why Use Approximation? (2 reasons)

Order-of-Magnitude Estimates

An order-of-magnitude estimate is a rough approximation based on powers of _____.

Example: mass of a car $\approx 10^{\text{_____}}$ kg

Example: number of people in a stadium $\approx 10^{\text{_____}}$

Worked Example: Breaths in a Lifetime

Without a calculator, estimate how many breaths an average person takes in their lifetime.

Breaths per minute (estimate):

Minutes per hour $\times$ hours per day (estimate):

Days per year:

Years in an average lifetime:

Multiply all factors — order-of-magnitude answer:

SUMMARY — in your own words, summarize this section in 2–3 sentences:

Chapter 2: One-Dimensional Kinematics

2.1 Displacement

CUE / KEY QUESTIONS

Define displacement

How is displacement different from distance?

Formula for displacement

Can displacement be zero when distance isn't? Example

What Is Displacement?

Displacement is a _____ quantity — it has both magnitude and _____.

Distance is a _____ quantity — it only tells you how much ground was covered.

Formula

$\Delta x = \text{_____} - \text{_____}$ (xf = final position, xi = initial position)

Worked Example

An object moves from xi = 2 m to xf = 5 m. Find the displacement.

$\Delta x =$

Distance vs. Displacement — fill in the table

Distance — Type: _____ (vector/scalar). Description:

Displacement — Type: _____ (vector/scalar). Description:

Direction Matters

Positive displacement = movement in the _____ direction.

Negative displacement = movement in the _____ direction.

SUMMARY — in your own words, summarize this section in 2-3 sentences:

2.2 Vectors, Scalars, and Coordinate Systems

CUE / KEY QUESTIONS

Scalar vs. vector — key difference

3 examples of each

2 methods of vector addition

3 rules for choosing a coordinate system

Scalars vs. Vectors

Scalar quantities have _____ only. Examples: _____, _____, _____

Vector quantities have both _____ and _____.
Examples: _____, _____, _____

Representing Vectors

Length of the arrow = _____. Direction of the arrow = _____.

Vector Addition

Two methods: the _____ method (tip-to-tail) and the _____ method (using components).

Example: A = 3 m east, B = 4 m north. Find the resultant R using the Pythagorean theorem.

$$R = \sqrt{A^2 + B^2} =$$

Coordinate Systems — 3 rules for choosing one

Components of a Vector

A vector V at angle θ breaks into: $V_x =$ _____ $V_y =$ _____

SUMMARY — in your own words, summarize this section in 2–3 sentences:

2.3 Time, Velocity, and Speed

CUE / KEY QUESTIONS

SI unit of time

Speed formula (scalar)

Velocity formula (vector)

Average vs. instantaneous velocity — difference?

Time

SI unit of time: _____

Speed vs. Velocity

Speed is a _____ quantity. Formula: speed = _____ / _____

Velocity is a _____ quantity. Formula: $v = \Delta x / \Delta t$, where $\Delta x =$ _____ and $\Delta t =$ _____

Average vs. Instantaneous Velocity

Average velocity: $v_{\text{avg}} = (\text{_____} - \text{_____}) / (\text{_____} - \text{_____})$

Instantaneous velocity is found using _____ — it's the slope of the _____ graph at a single point.

Worked Example

A car travels 30 mi/h for half the trip time and 50 mi/h for the other half. Is the average speed 40 mi/h, more, or less?

Your prediction:

Calculation:

Actual average speed:

SUMMARY — in your own words, summarize this section in 2–3 sentences:

2.4 Acceleration

CUE / KEY QUESTIONS

Define acceleration

Formula for average acceleration

Units of acceleration

Positive vs. negative acceleration — meaning?

What Is Acceleration?

Acceleration is the rate at which _____ changes over time. It is a _____ quantity.

An object accelerates if it (list 3 ways):

Formula for Average Acceleration

$a = (\text{_____} - \text{_____}) / \text{_____}$ where $v_f = \text{_____}$, $v_i = \text{_____}$, $t = \text{_____}$

Units of acceleration: _____

Positive vs. Negative Acceleration

Positive acceleration: velocity increases in the _____ direction.

Negative acceleration (deceleration): velocity _____.

Real-World Example

Gravity causes objects to accelerate downward at $g \approx \text{_____}$ m/s^2 .

SUMMARY — in your own words, summarize this section in 2–3 sentences:

2.5 Motion Equations for Constant Acceleration

CUE / KEY QUESTIONS

What does 'constant acceleration' mean?

Write all 4 kinematic equations from memory

Which variable is missing from each equation?

What Is Constant Acceleration?

Constant acceleration means velocity changes by the _____ amount in each unit of time.

The Four Kinematic Equations — fill in each

1. Displacement with average velocity:
2. Velocity as a function of time:
3. Displacement as a function of time:
4. Velocity as a function of displacement:

Which variable is missing from each equation above? List all 4:

Worked Example

A car starts from rest and accelerates at 2 m/s^2 for 5 seconds. How far does it travel?

Given:

Find:

Equation to use:

Solve:

Answer (with units):

SUMMARY — in your own words, summarize this section in 2-3 sentences:

2.6 Problem-Solving Basics

CUE / KEY QUESTIONS

List all 9 steps of the problem-solving strategy, in order

General Problem-Solving Strategy — fill in all 9 steps, in order

SUMMARY — in your own words, summarize this section in 2-3 sentences:

2.7 Falling Objects

CUE / KEY QUESTIONS

What is free fall?

Value and direction of g

Who first showed all objects fall at the same rate?

Sign convention for free-fall problems

Free Fall and Gravity

Free fall is motion under the influence of _____ alone, with no _____.

All objects in free fall near Earth's surface accelerate downward at $g =$ _____ m/s^2 .

_____ first demonstrated that all objects fall at the same rate, regardless of mass (no air resistance).

Direction and Sign Conventions

Choose _____ or _____ as the positive direction, and stay consistent.

Worked Example 1: Object Dropped from Rest

A ball is dropped from a height of 20 m. How long does it take to hit the ground?

Given:

Equation:

Solve:

Answer:

Worked Example 2: Object Thrown Upward

A ball is thrown straight up at 15 m/s. How high does it go before its velocity is momentarily zero?

Given:

Equation:

Solve:

Answer:

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SUMMARY — in your own words, summarize this section in 2-3 sentences:

Chapter 3: Two-Dimensional Kinematics

3.1 Kinematics in Two Dimensions: An Introduction

CUE / KEY QUESTIONS

Key principle:
independence of
perpendicular motions

9 blocks east + 5 blocks
north — straight-line
distance?

Why do we resolve
motion into
components?

What Is Two-Dimensional Motion?

Two-dimensional motion involves movement along _____
perpendicular axes — typically _____ (horizontal) and _____ (vertical).

Key Principle: Independence of Perpendicular Motions

Motion in one direction does _____ affect motion in the other
direction.

Worked Example: Walking in a City

Walk 9 blocks east, then 5 blocks north. Find the straight-line distance using
the Pythagorean theorem.

Straight-line distance =

Dropped vs. Thrown Ball (same height, no air resistance)

Vertical speed comparison:

Horizontal speed comparison:

Conclusion:

SUMMARY — in your own words, summarize this section in 2–3 sentences:

3.2 Vector Addition and Subtraction: Graphical Methods

CUE / KEY QUESTIONS

Describe the head-to-tail method

Is vector addition commutative?

How do you subtract vectors graphically?

Effect of multiplying a vector by a negative scalar?

Head-to-Tail Addition

To add $A + B$: draw A, then draw B starting at the _____ of A. The resultant runs from the _____ of A to the _____ of B.

$A + B$ _____ $B + A$ (write = or $\neq$ — is vector addition commutative?)

Vector Subtraction

$A - B = A + (\text{_____})$

Graphically: flip B by _____° before adding head-to-tail.

Scalar Multiplication

$2A$ is _____ as long as A, _____ direction.

$-0.5A$ is _____ as long as A, _____ direction.

Worked Example

A hiker walks 27.5 m east, then 25.0 m north. Sketch to scale and estimate R and θ using a ruler and protractor.

R (estimated) =

θ (estimated) =

Limits of the Graphical Method

Accuracy is limited by _____.

SUMMARY — in your own words, summarize this section in 2-3 sentences:

3.3 Vector Addition and Subtraction: Analytical Methods

CUE / KEY QUESTIONS

Component formulas: $A_x = ?$
 $A_y = ?$

4-step process for adding
vectors analytically

Formulas for magnitude and
direction of the resultant

Resolving a Vector into Components

For vector A at angle θ from the x-axis: $A_x = \underline{\hspace{2cm}}$ $A_y = \underline{\hspace{2cm}}$

$$A = \sqrt{(\underline{\hspace{1cm}} + \underline{\hspace{1cm}})}$$

Adding Vectors by Components — fill in the 4 steps

Worked Example: Hiker Revisited

Same hiker: 27.5 m east + 25.0 m north. Solve analytically.

$$R_x =$$

$$R_y =$$

$$R =$$

$$\theta =$$

Worked Example: Non-Axis-Aligned Vector

A 30.0 m displacement at 45° north of east. Find the x- and y-components.

$$C_x =$$

$$C_y =$$

SUMMARY — in your own words, summarize this section in 2–3 sentences:

3.4 Projectile Motion

CUE / KEY QUESTIONS

2 assumptions of basic projectile motion

$a_x = ?$ $a_y = ?$ (y-axis pointing up)

v_{0x} and v_{0y} in terms of v_0 and θ

Formula for range R

What Is Projectile Motion?

Projectile motion is the motion of an object thrown into the air, subject only to _____.

Horizontal motion: _____ velocity (no acceleration). Vertical motion: _____ accelerated motion due to gravity.

Coordinate Setup

If the y-axis points up: $a_x = \underline{\hspace{2cm}}$ $a_y = \underline{\hspace{2cm}}$

Initial Velocity Components

$v_{0x} = \underline{\hspace{2cm}}$ $v_{0y} = \underline{\hspace{2cm}}$

Kinematic Equations for Projectile Motion — fill in (x and y separately)

Horizontal:

Vertical:

Worked Example: Fireworks Shell

A shell is shot at 70.0 m/s at 75.0° above horizontal. Find (a) max height, (b) time to reach it, (c) horizontal displacement at that time.

$v_{0x} =$

$v_{0y} =$

(a) Height at highest point =

(b) Time to highest point =

(c) Horizontal displacement =

Worked Example: Zircon Rock Toss

An astronaut tosses a rock horizontally at 6.95 m/s on planet Zircon. It falls 1.40 m and lands 8.75 m away. Find g on Zircon.

Given:

Time to land, $t =$

g on Zircon =

Range Formula

$R =$ _____ (in terms of v_0 , g , θ) — maximum range occurs at θ
= _____°

SUMMARY — in your own words, summarize this section in 2–3 sentences: